

RADIUS & TACACS+

Task 1 — Create a comparison table listing at least 4 key differences between RADIUS and TACACS+ features, focusing on protocol type, encryption, ports used, and command authorization.

Feature	RADIUS (Remote Authentication Dial-In User Service)	TACACS+
Protocol Type	Open industry standard , supported by almost every network vendor.	Originally Cisco-proprietary; still mainly used on Cisco (and Cisco-compatible) gear.
Transport	UDP — connectionless, faster, but no delivery guarantee.	TCP — connection-oriented, reliable delivery, built-in acknowledgement.
Encryption	Encrypts only the password field in the Access-Request packet; the rest of the packet (username, attributes) is sent in clear text.	Encrypts the entire packet body, including username and all attributes — only the header is unencrypted.
Ports Used	UDP 1812 (authentication) and 1813 (accounting).	TCP port 49 for all AAA traffic.
Command Authorization	No native per-command control — authorization is usually limited to broad access levels (VLAN, ACL, session attributes).	Supports command-level authorization — can allow or block individual CLI commands per user or privilege level (e.g. allow "show" but block "reload").
Typical Use Case	Network access control — WiFi logins, VPN, 802.1X wired/wireless authentication.	Device administration — controlling and auditing admin/engineer logins to routers and switches.

Task 2 — Draw a simple flowchart or sequence diagram that illustrates the RADIUS authentication process for a user trying to log in to a WiFi network (like at a coffee shop or college campus).

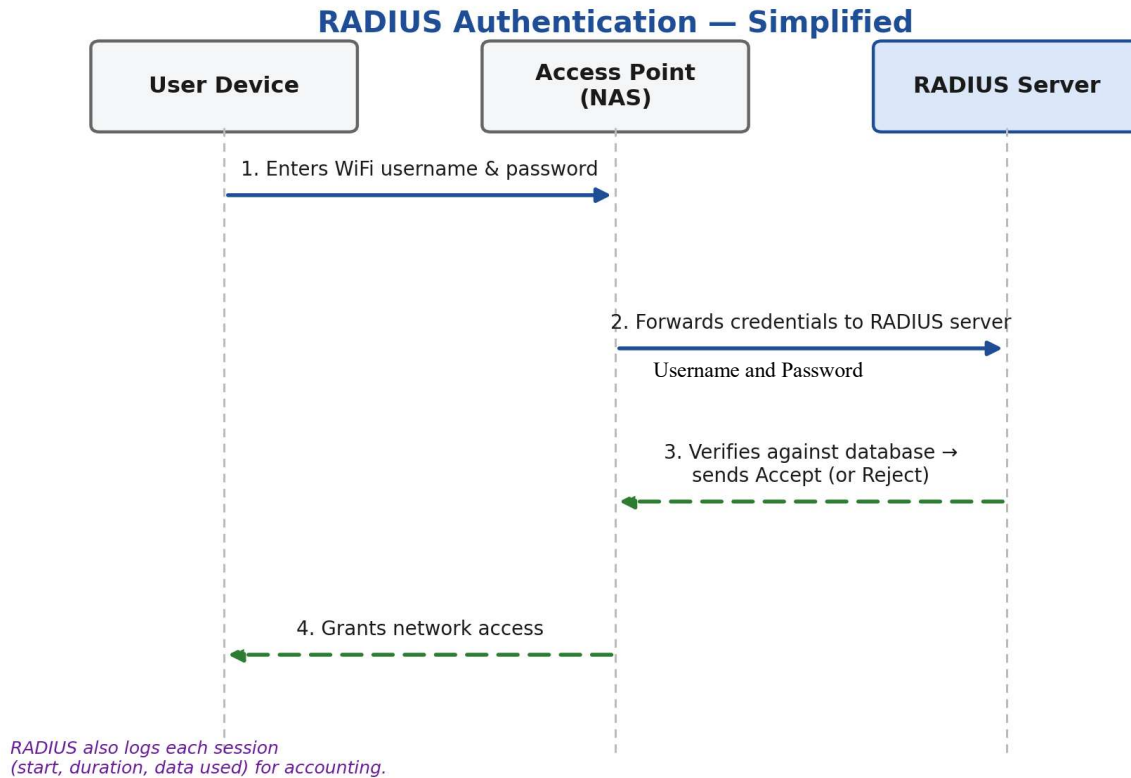


Figure 1: The Access Point (NAS) forwards the login request to the RADIUS server, which checks the user database, returns Accept/Reject with access attributes, and logs the session via Accounting-Start/Stop.

Task 3 — Describe a real-world scenario from any app you use (such as Zomato, Paytm, or IRCTC) where TACACS+ would be more suitable than RADIUS for managing admin access, and explain why.

Zomato's network team manages core routers and switches across multiple data centers that carry sensitive data — customer addresses, payment details, and restaurant partner information. When a network engineer logs in to configure one of these devices, TACACS+ is the better fit for a few reasons.

First, command-level control matters here: a junior engineer might be authorized to run diagnostic commands like "show interface" or "show running-config", but should be blocked from destructive commands like "reload" or "erase startup-config". TACACS+ can enforce this per command; RADIUS can't.

Second, full-packet encryption protects the entire admin session (not just the password) — important since these sessions can include configuration details about the internal network.

Third, TACACS+'s separate accounting process logs every single command each engineer runs, which gives Zomato a precise audit trail if a mis-configuration or unauthorized change causes an outage — useful for both internal review and compliance around handling sensitive user data. RADIUS, by contrast, is a better fit for Zomato's guest/office WiFi or VPN login, where broad accept/reject decisions are enough and speed matters more than command-level granularity.

Task 4 — List two advantages and two disadvantages of using RADIUS for network authentication in a public hotspot (like those found in malls or airports).

Advantages:

1. Wide compatibility — RADIUS is an open standard which is supported virtually by every access point, router, so a mall or airport can mix hardware from different manufacturers and still centralize logins on one RADIUS server.
2. Fast, lightweight authentication — because it runs over UDP with a simple request/response exchange, RADIUS handles quick connect/disconnect cycles well, which suits a hotspot with thousands of different devices joining and leaving throughout the day.

Disadvantages:

1. Limited encryption — RADIUS only encrypts the password field; other attributes in the packet (like username or assigned access rights) travel unencrypted, which is a real risk on a public network where traffic could be intercepted.
2. Unreliable transport — UDP doesn't guarantee delivery, so on a congested public WiFi network with heavy simultaneous usage, authentication requests can be dropped or delayed, occasionally causing failed or slow logins.

Task 5 — Use ChatGPT or another AI tool to generate a sample TACACS+ configuration for a Cisco router, then review the output and highlight any commands you would modify for a college lab setup.

```
! Sample TACACS+ configuration for a Cisco router
!
aaa new-model
!
tacacs-server host 192.168.1.10 key MySecretKey123
!
aaa authentication login default group tacacs+ local
aaa authorization exec default group tacacs+ local
aaa authorization commands 15 default group tacacs+ local
aaa accounting exec default start-stop group tacacs+
aaa accounting commands 15 default start-stop group tacacs+
!
username backupadmin privilege 15 secret LocalBackupPass1
!
line vty 0 4
  login authentication default
  transport input ssh
!
```